Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/28/2008
PI Name:	Warchol, Mark Edward Ph.D.		
Project Title:	Patterning of the Vestibular Sensory Organs		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	warcholm@ent.wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(314) 747-7186
Organization Name:	Washington University		
PI Address 1:	Dept. of Otolaryngology, Box 8115		
PI Address 2:	660 South Euclid Ave.		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63110	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per A. Chu (4/07) NOTE: Received NCE to 3/31/2007, per A. Chu (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project focuses on identification of the cellular and molecular mechanisms that regulate the development of the utricle, the principal 'gravity detector' of the vertebrate inner ear. To characterize the expression patterns of the GATA3 and PAX2 transcription factors during the embryonic development of the utricle and determine the relationship between the expression of these factors and key morphogenic events. The vestibular (balance) sensory organs of the human inner ear contain two type of sensory cells, but the specific roles of these cells are not understood. Knowledge gained from these studies should reveal developmental differences between these two classes of sensory cell as well as suggest how they develop their proper orientation. Both of these processes are crucial for normal balance function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge about the development and maintenance of the vestibular sensory organs could lead to new methods for the diagnosis of and treatments for balance disorders
Task Progress:	The aim of this project was to characterize the expression patterns of the GATA3 and PAX2 transcription factors in the developing chick vestibular organs and to investigate the role of these factors in inner ear morphogenesis. During the funding period of this project, that aim has been accomplished. We have used immunohistochemistry to profile the expression of GATA3 and PAX2 in the developing inner ear of chick embryos. Based on those results, we also attempted to alter the expression of GATA3, by treating the developing ears with the morphogen sonic hedgehog (SHH). Notably, exogenous SHH caused profound changes in the structure of the developing vestibular system, suggesting that this molecule plays a key role in vestibular patterning.
Bibliography Type:	Description: (Last Updated: 03/19/2008)
Abstracts for Journals and Proceedings	Drescher AJ, Kach J, Warchol ME. "Ectopic application of sonic hedgehog in the developing chick inner ear leads to reduced GATA3 expression and disorganization of the vestibular organs." Association for Research in Otolaryngology Midwinter Research Meeting, Baltimore, MD, February 2006. Association for Research in Otolaryngology Midwinter Research Meeting, Baltimore, MD, February 2006. Abstract #651. , Feb-2006
Abstracts for Journals and Proceedings	Drescher AJ, Warchol ME. "Expression Patterns of GATA3 in the Developing Chick Inner Ear." Association for Research in Otolaryngology Midwinter Research Meeting, New Orleans, LA, February 2005. Association for Research in Otolaryngology Midwinter Research Meeting, New Orleans, LA, February 2005. Abstract #432. , Feb-2005
Articles in Peer-reviewed Journals	Drescher AJ, Kach J, Warchol ME. "Sonic hedgehog influences GATA3 expression and formation of the vestibular organs in the developing chick inner ear." Manuscript in preparation, February 2008. , Feb-2008